Patrick Cherry, PhD

patrick [at] pcherry [dot] org | upon request | Principal Scientist | Oncology | BillionToOne | San Francisco, California | pcherry.org | github.com/pdcherry | linkedin.com/in/p-cherry | United States Citizen | Updated on 2026-10-19

PhD data scientist and professional tool-builder and science communicator. I bring the incisive critical thinking of an -omics-trained scientist to data science. Skilled at data visualization, pipeline building, and custom tool building and validation. I am dedicated to serving as the bridge between complex data problems and action-oriented leadership seeking data-driven strategy.

Industry Experience

Current | Oct 2026

Principal Scientist, Oncology

BillionToOne, Inc. — Menlo Park, California

Oct 2026 | Sep 2024

Senior Scientist, Oncology

BillionToOne, Inc. — Menlo Park, California

- Tech lead and program manager for multi-quarter cfDNA genomics assay expansion (Select v3); led the launch and CAP/CLIA validation of Northstar Select Pharmacogenomics (PGx) in under two months
- Implemented department-wide copy number database using extract-transform-load (ETL) procedure on production data
- Ideated, coded, and validated patented method for tumor fraction estimation from plasma aneuploidy data using Gaussian mixture models, custom model selection metric, and custom consensus logic
- Answered business and product design questions from leadership about un-met customer need with internal reports on Notion written in reproducible coding notebooks using Python and R in a HIPAA-compliant environment
- As hiring manager, recruited and hired four Research Associates; managed the Statistics interview

Apr 2024 | Dec 2021

Senior Scientist, Genomics

Twist Bioscience, Inc. — South San Francisco, California

- Invented and introduced multiple new products to market, yielding $1M+ in new revenue as *Tech Lead*, including: *Pan-cancer cfDNA* (v1 & v2) (patented), *CNV Controls*, *RNA Fusion Controls*, RNA-seq, and patented adapter removal
- Answered feasibility questions with Design of Experiment principles and data analysis using `R` and `Python`.
- Wrote, documented, and maintained production QC pipelines for sub-components and finished good QC; trained end-users in Manufacturing and Quality on their use, interpretation, and reporting.
- Maintained production code (`git` version control with code review and team best practices) with CD (Github actions); operated in AWS & Databricks high-performance compute environment using Unix shell
- Answered manufacturing & business questions using SQL / Snowflake database of production events and customer data

Jan 2021 | Apr 2019

Scientist I, Next Generation Sequencing (NGS)

Zymergen, Inc. — Emeryville, California

- Boosted pan-project NGS genotyping success by 45% using Design of Experiment and Estimated Marginal Means modeling for causal inference; Guided demanding and diverse internal customers on complex NGS experiments
- Built and disseminated reproducible statistical templates in `Rmarkdown` and `JMP` for self-serve NGS data exploration
- Rendered strain build and plasmid build reports from LIMS `SQL` database in `MariaDB` and R / `ggplot2`
- Designed, built, and demonstrated a patented counterselectable marker for the non-model microbe *Bacillus subtlis*

Education

Apr 2019 | Aug 2013

PhD in Molecular Biology

University of Colorado School of Medicine — Aurora/Denver, Colorado

- Ph.D. in Molecular Biology. Advisor: Jay Hesselberth, PhD. Key courses: 6606 Statistics; 7621 Genome Analysis Workshop.
- Thesis: RNA terminus chemistry potentiates decay events targeting *HAC1* mRNA during the Unfolded Protein Response

May 2013 | Aug 2009

BA in Molecular Biology with Distinction, minor in Mathematics

Hendrix College — Conway, Arkansas

- Senior Capstone Project showing changes of key stress response gene expression to oxidative stress in liver cells
- Key courses: Probability & Statistics; Discrete Mathematics; Adv Biological Chemistry, Analytical Chemistry